- DISTRIBUZIONE

$$\mathcal{D}(\Omega) = C_c^\infty(\Omega)$$

$$\mathcal{D}^*(\Omega)$$

$$\lim_{n\to\infty} T_n(\varphi) = T(\varphi) \quad \forall \varphi \in \mathcal{D}(\Omega)$$

$$L^2(\Omega) := \left\{ f : \Omega \to \mathbb{R} \ : \int_\Omega |f|^2 < \infty \right\} \Big/ \sim$$

$$f \sim g \qquad f = g \quad q.o.$$

$L^2(\Omega)$ è HILBERT $\quad \langle f, g \rangle := \int_\Omega f g \, dx$

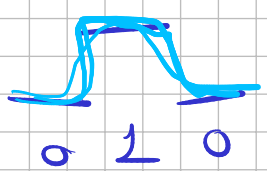
$$\|f\|_{L^2} = \sqrt{\int_\Omega f^2 \, dx}$$

$$L^2(\Omega) = (L^2(\Omega))^*$$

$$f \to T_f \qquad T_f(g) := \int_\Omega f g \, dx \quad \forall g \in L^2$$

$$f_T \leftarrow T \qquad T(g) = \int_\Omega f_T \cdot g \, dx \quad \forall g \in L^2$$

$\mathcal{D}(\Omega)$ SONO IN DENSE IN L^2



$$\forall f \in L^2 \quad \exists \varphi_k \in \mathcal{D}(\Omega) : \|\varphi_k - f\|_{L^2} \xrightarrow[k\to\infty]{} 0$$

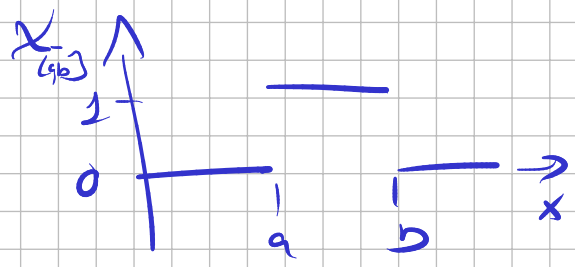
$$\underbrace{D(\Omega)} \subset \underbrace{L^2(\Omega)} \overset{\Downarrow}{\cong} \underbrace{(L^2(\Omega))^*} \subset \underbrace{D^*(\Omega)}$$

Se $\exists\, F \in (L^2(\Omega))^*$

F è LINITATO $\forall\, \varphi \in D(\Omega)$

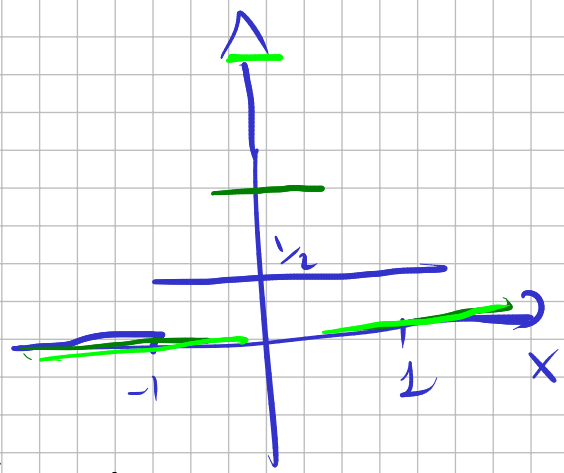
$F \in D^*(\Omega)$

$$\chi_{[a,b]}(x) = \begin{cases} 1 & x \in [a,b] \\ 0 & x \notin [a,b] \end{cases}$$



$\chi_{[a,b]} \in L^2(\mathbb{R})$

$$f_n(x) := \frac{n}{2}\, \chi_{\left[-\frac{1}{n}, \frac{1}{n}\right]}$$



$$\int f_n(x) = \frac{n}{2} \int_{-\frac{1}{n}}^{\frac{1}{n}} 1\, dx = \frac{n}{2}\left(\frac{2}{n}\right) = 1$$

$\forall\, \varphi \in D$ C^∞_c

$$T_{f_n}(\varphi) = \int_{\mathbb{R}} f_n \cdot \varphi\, dx = \frac{n}{2} \int_{-\frac{1}{n}}^{\frac{1}{n}} \varphi(x)\, dx$$

$$= \frac{n}{2}\left[\Phi\left(\frac{1}{n}\right) - \Phi\left(-\frac{1}{n}\right)\right]$$

$$\frac{d}{dx}\Phi(x) = \varphi(x)$$

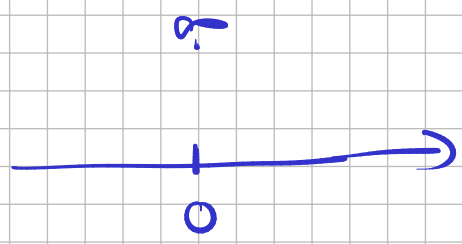
$$h_n = \frac{1}{n} \qquad T_{f_n}(\varphi) = \frac{\Phi\left(\frac{1}{n}\right) - \Phi\left(-\frac{1}{n}\right)}{2h_n}$$

$$\lim_{n\to\infty} \underbrace{T_{f_n}(\varphi)} = \lim_{n\to\infty} \frac{\Phi(h_n) - \Phi(-h_n)}{2h_n}$$

$$= \frac{d\Phi(0)}{dx} = \underbrace{\varphi(0)} = \delta_0(\varphi)$$

$$T_{f_n} \to \delta_0 \in \mathcal{D}^*$$

$$\notin L^2$$



DEFINIZIONE DERIVATA DI DISTRIB.

$$T \in \mathcal{D}^*(\Omega)$$

$$\partial_{x_i} T \in \mathcal{D}^* \qquad \varphi \in \mathcal{D}(\Omega)$$

$$\partial_{x_i} T(\varphi) = \int_\Omega \overbrace{\partial_{x_i} T(x)}^{\text{"}\quad\text{"}} \cdot \varphi(x)\, dx$$

$$\underbrace{\int_{\partial\Omega} T(x) \cdot \underbrace{\varphi(x)}_{\in C_c^\infty(\Omega)}\, ds}_{=0} - \int_\Omega T(x) \cdot \partial_{x_i} \varphi(x)$$

$$= -\langle T, \partial_{x_i} \varphi(x) \rangle$$

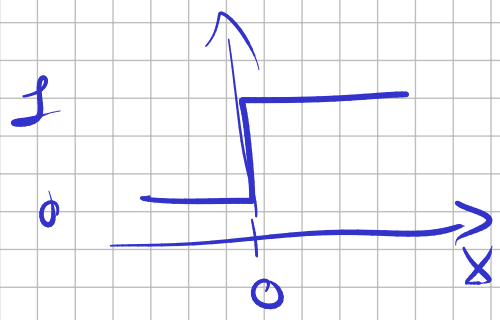
$$\boxed{\partial_{x_i} T(\varphi) := -T(\partial_{x_i}\varphi)}$$

$$\left\langle \frac{\partial^{p_1+\dots+p_d} T}{\partial x_1^{p_1} \dots \partial x_d^{p_d}}, \varphi \right\rangle := (-1)^{p_1+\dots+p_d} \left\langle T, \frac{\partial^{p_1+\dots+p_d}\varphi}{\partial x_1^{p_1} \dots \partial x_d^{p_d}} \right\rangle$$

$$T(\varphi) =: \langle T, \varphi \rangle$$

$\Omega = [-1, 1]$

$$R(x) = \begin{cases} 1 & x > 0 \\ 0 & x \leq 0 \end{cases}$$



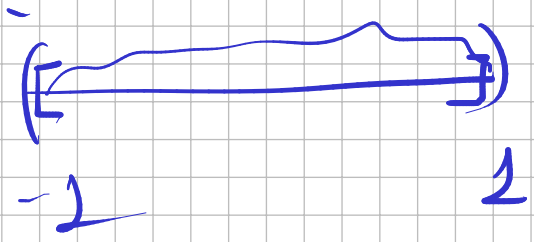
$$\langle \partial_x T_R, \varphi \rangle = -\int_{-1}^{1} R(x) \cdot \partial_x \varphi \, dx$$

$$= -\int_0^1 \partial_x \varphi(x)\, dx = -[\varphi]_0^1 =$$

$$= -\varphi(1) + \varphi(0) = \varphi(0) = \delta_0(\varphi)$$

$\varphi \in \mathcal{C}_c^\infty$ $\quad \varphi(1) = 0$

$$\partial_x T_R = \delta_0$$



SPAZI DI SOBOLEV

$(L^2)^* \subset \mathcal{D}^*(\Omega)$

$\Omega \subset \mathbb{R}^d \qquad k \in \mathbb{N}_0$

$$H^k(\Omega) := \{ f \in L^2 : \partial_{x_1^{p_1}} \cdots \partial_{x_d^{p_d}} f \in L^2 \quad \forall\, p_1 + \dots + p_d \leq k \}$$

$H^{k+1}(\Omega) \subset H^k(\Omega)$

$H^0(\Omega) = L^2(\Omega)$

- es. HEAVISIDE $R(x) = \begin{cases} 1 & x > 0 \\ 0 & x \leq 0 \end{cases}$

$R \in L^2 \qquad R' = \delta_0 \notin L^2$

$R \notin H^1$

- es $|x|$ ✓ $\qquad |x|' =$

$|x|'' \approx \delta_0$

$|x| \in H^1 \qquad |x| \notin H^2$

es. $f(x) = \begin{cases} x & x > 0 \\ 0 & x \leq 0 \end{cases}$

$f'(x) = \begin{cases} 1 & x > 0 \\ 0 & x \leq 0 \end{cases}$ $\qquad f' \in L^2$

$f'' = \delta_0 \qquad \Rightarrow f'' \notin L^2((-1,1))$

$f \in H^1 \qquad f \notin H^2$

$\langle f, g \rangle_{L^2} = \int f \cdot g \, dx$

$$(f,g)_{H^k(\Omega)} := \sum_{p_1 + \dots p_d \leq k} \int_\Omega \partial_{x_1^{p_1} \dots x_d^{p_d}} f \cdot \partial_{x_1^{p_1} \dots x_d^{p_d}} g \, dx.$$

$$\|f\|_k = \sqrt{(f,f)_{H^k(\Omega)}} := \sqrt{\sum_{p_1 + \dots + p_d \leq k} \int_\Omega (\partial_{x_1^{p_1} \dots x_d^{p_d}} f)^2 \, dx}$$

$$|f|_k = |f|_{H^k(\Omega)} := \sqrt{\sum_{p_1 + \dots + p_d = k} \int_\Omega (\partial_{x_1^{p_1} \dots x_d^{p_d}} f)^2 \, dx}$$

$|f|_k = 0 \qquad f \equiv 1 \qquad |f|_1 = 0 \qquad \Omega = (-1,1)$

- $\|f\|_k = \sqrt{\sum_{m=0}^{k} |f|_m^2}$

$\Omega = (-1,1) \subset \mathbb{R}$ $\qquad f, g \in H^1(\Omega)$

$(f,g)_{H^1} = (f,g)_1 = \int_\Omega f g \, dx + \int_\Omega f' \cdot g' \, dx$

$\|f\|_1 = \sqrt{\int_\Omega f^2 \, dx + \int_\Omega (f')^2 \, dx}$

$|f|_1 = \sqrt{\int_\Omega (f')^2 \, dx} < \infty$

<u>TEOREMA</u>

$\Omega \subset \mathbb{R}^d$ APERTO e LIMITATO

$\partial\Omega$ LISCIO $\Rightarrow H^k(\Omega) \subset \mathcal{C}^m(\overline{\Omega})$

se $m < k - \frac{d}{2}$.

<u>ESEMPI</u> $\qquad d = 1 \qquad m < k - \frac{1}{2}$

es. $\quad k = 1 \qquad H^1((-1,1)) \subset \mathcal{C}^0([-1,1])$

$0 < 1 - \frac{1}{2} = \frac{1}{2}$ ✓

$z(x) \qquad z' \in L^2 \qquad z \in H^1((-1,1))$

$z(x) \in C^0([-1,1])$

$d=2 \qquad m < k - \frac{2}{2} = k-1$

$H^1((-1,1)^2) \not\subset C^0([-1,1]^2)$

$H^2((-1,1)^2) \subset C^0([-1,1]^2)$

$k=2 \qquad m=0 \qquad 0 < 2-1 = 1$ ✓

$\underbrace{H^1_0(\Omega)}_{\text{"}} = \overline{\underbrace{D(\Omega)}_{C_c^\infty(\Omega)}}^{H^1(\Omega)}$

"FUNZIONI IN H^1 CHE SONO ABBASTANZA ZERO AL BORDO"

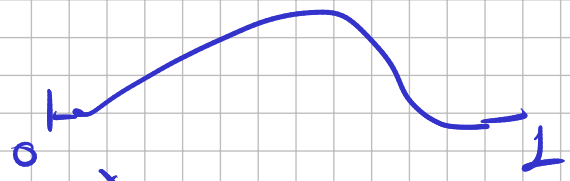
TEOREMA: DISUGUAGLIANZA DI POINCARÉ

$\Omega \subset \mathbb{R}^d$ LIMITATO $\quad \partial\Omega$ LIPSCHITZ

$\exists\, C = C(\Omega) > 0 : \quad \forall\, u \in H^1_0(\Omega)$

$$\|u\|_{L^2(\Omega)} \le C \|\nabla u\|_{L^2(\Omega)} = C |u|_{H^1(\Omega)}$$

es. 1D per C_c^∞

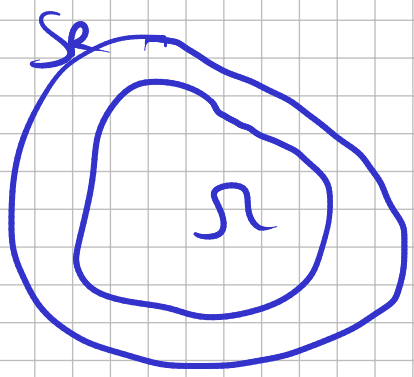


$$u(x) = \int_0^x u'(\xi)\, d\xi$$

$$\|u\|^2 \le C(\Omega) \|u'\|^2$$

DIMOSTRAZIONE $\Omega \subset \mathbb{R}^d$ LIMITATO

$S_R = \{ x : |x - x_0| < R \}$



$D(\Omega)$ è denso in H_0^1

$$\operatorname{div}(x - x_0) = d$$

$$\operatorname{div} \begin{pmatrix} x_1 - x_1^0 \\ x_2 - x_2^0 \\ \vdots \\ x_d - x_d^0 \end{pmatrix} = \begin{matrix} \partial_{x_1}(x_1 - x_{10}) \\ + \partial_{x_2}(x_2 - x_{02}) \\ \vdots \\ + \partial_{x_d}(x_d - x_{0d}) \end{matrix} = \underbrace{1 + 1 \dots + 1}_{= d}$$

$$\|u\|^2_{L^2(\Omega)} = d^{-1} \int_\Omega d\, |u(x)|^2 dx$$

$$= d^{-1} \int_\Omega \operatorname{div}(x - x_0) \cdot |u(x)|^2 dx$$

$$= d^{-1} \int_{\partial\Omega} \underbrace{u(x)}_{=0} \cdot (x - x_0) \cdot n \cdot dS \Big\} = 0$$

$$- d^{-1} \int_\Omega (x-x_0) \cdot \nabla \left(u(x)^2 \right) dx$$

$$= -2d^{-1} \int_\Omega u(x) \cdot \underbrace{(x-x_0)}_{\leq R} \nabla u(x) \, dx$$

$$\leq R \left| 2d^{-1} \int_\Omega u(x) \cdot \nabla u(x) \, dx \right|$$

$$\leq 2Rd^{-1} \sqrt{\int_\Omega u^2 \, dx} \sqrt{\int_\Omega (\nabla u)^2 \, dx}$$

$$\leq \overbrace{2Rd^{-1}}^{C} \|u\|_{L^2} \|\nabla u\|_{L^2}$$

$$\|u\|_{L^2}^2 \leq C \|u\|_{L^2} \cdot \|\nabla u\|_{L^2}$$

$$\|u\|_{L^2} \leq C \|\nabla u\|_{L^2} \leq C |u|_1$$

$$\|u\|_1^2 = |u|_1^2 + \|u\|_{L^2}^2 \leq$$

$$\leq |u|_1^2 + C^2 |u|_1^2 = (1+C^2) |u|_1^2$$

PDE ELLITTICHE LINEARI

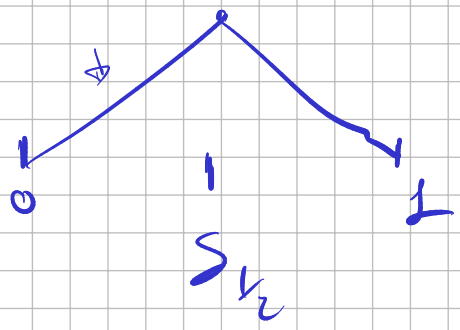
$\Omega \subset \mathbb{R}^d \quad d > 1 \qquad u : \Omega \to \mathbb{R}$

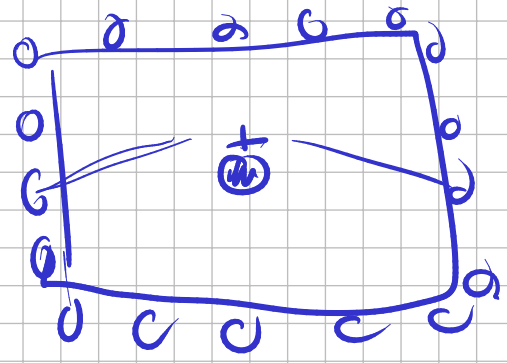
$$-\Delta u = f \qquad \forall x \in \Omega \qquad + \text{B.C.}$$

$$\Delta u = \partial^2_{x_1} u + \dots + \partial^2_{x_d} u$$

$f \equiv 0$ CASO OMOGENEO

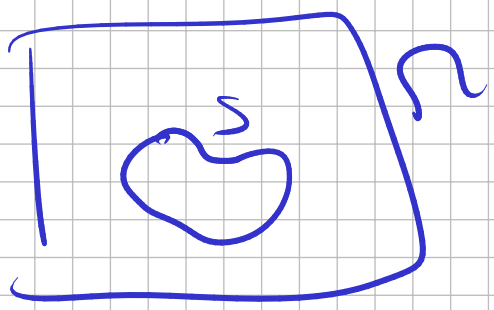
$-\Delta u = 0$ EQUAZIONE DI LAPLACE





u TEMPERATURA

$S \subset \Omega$



$$\int_{\partial S} F \cdot n \, d\Gamma = 0$$

$F \sim -\nabla u \qquad F = -a \nabla u$

$$0 = \int_S \operatorname{div}(F) \, dx \qquad \forall S \subset \Omega$$

$$\Rightarrow \operatorname{div}(F(u)) = 0 \qquad \forall x \in \Omega$$

$$\operatorname{div}(F(u)) = \operatorname{div}(-a \nabla u) = -a \operatorname{div}(\nabla u)$$

$$-a\,\Delta u = 0$$

$$\operatorname{div}\begin{pmatrix} \partial_{x_1} u \\ \vdots \\ \partial_{x_d} u \end{pmatrix} = \partial_{x_1}\partial_{x_1} u + \partial_{x_2}\partial_{x_2} u + \dots + \partial_{x_d}\partial_{x_d} u =$$

$$\partial_{x_1}^2 u + \dots + \partial_{x_d}^2 u = \Delta u .$$